

Individual contribution statement

CSV Data Quality Triage Agent | Lab 9 Phase 3

Parthiv Godrihal

Roll number I024 **Responsibility** Agent development and application integration

My contribution focused on the LLM agent workflow, LangChain integration and the Streamlit application. I worked on connecting the model, diagnostic tools and reporting interface into a usable CSV quality assessment workflow.

Individual contributions

- Contributed to the LangChain agent setup, prompts, model configuration and tool integration. This workflow lets the Groq openai/gpt-oss-120b model select relevant diagnostics for the user question.
- Worked on the LCEL reporting chain and Pydantic output parsing so that the application produces a structured diagnosis while preserving verified tool findings and coverage limitations.
- Integrated the Streamlit workflow for CSV loading, target selection, diagnostic progress, visible tool traces and evidence download.
- Contributed to the reporting improvements that display an LLM summary, interpretation, suggested next steps and brief tool-choice explanations alongside the measured results.

Project evidence

Relevant project files include app.py, src/agent/factory.py, src/agent/prompt.py, src/agent/tools.py, src/agent/report_chain.py, src/models/schemas.py, src/ui/components.py and src/services/triage_service.py. Live traces and screenshots document the integrated application.

AI assistance

AI tools, including Codex, assisted with implementation, debugging, test execution, documentation and evidence capture. My contribution is described within this assisted team workflow; generated material and project evidence remain subject to team review.

Declaration

I confirm that this statement accurately describes my contribution to the project and the assistance used.

Signature _____ Date _____

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Nilay Jain

Roll number I029 **Responsibility** Data diagnostics and evaluation

My contribution focused on CSV diagnostics, dataset handling, testing and evaluation. I worked on the measurements and validation evidence used to assess whether the agent selected suitable checks and reported findings supported by tool output.

Individual contributions

- Contributed to deterministic checks for dataset profiling, missing values, duplicate rows, constant columns, high cardinality, outliers, class imbalance and strong correlation.
- Worked on dataset loading and backend evaluation, including the Pandas path for small CSVs and the DuckDB path for larger files, with explicit limits for expensive diagnostics.
- Contributed to sample datasets, expected findings, unit and integration tests, and checks for malformed inputs, failed tool calls and report validation.
- Evaluated scenario outputs and large-file benchmark records, including tool selection, finding accuracy, coverage, resource limits and retained failed attempts. This supported the documented 250 MB local target.

Project evidence

Relevant project files include `src/diagnostics/`, `src/data/`, `tests/`, `scripts/evaluate_phase3.py` and the benchmark scripts. Evaluation records in `docs/evaluation/phase3/` preserve scenario results, test output and resource measurements.

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